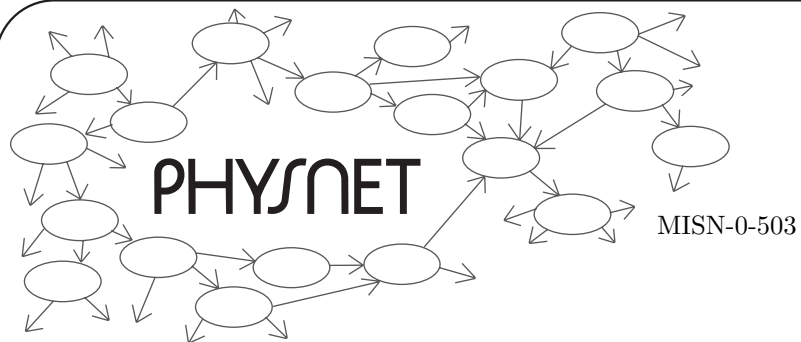




REVIEW OF VECTOR ANALYSIS

Electricity and Magnetism

Project PHYSNET • Physics Bldg. • Michigan State University • East Lansing, MI

REVIEW OF VECTOR ANALYSIS

by
R. D. Young

1. Introduction	1
1. Procedure	1
Acknowledgments	2

Title: **Review of Vector Analysis**

Author: R.D. Young, Dept. of Physics, Ill. State Univ.

Version: 10/18/2001

Evaluation: Stage B0

Length: 2 hr; 8 pages

Input Skills:

1. Vocabulary: Cartesian coordinates, normal (direction).
2. Take partial derivatives of simple functions.
3. Perform multiple integration on simple functions.
4. Expand any given simple function in a Taylor series (MISN-0-4).

Output Skills (Knowledge):

- K1. Vocabulary: scalar, vector, scalar field, vector field, vector addition and subtraction, multiplication of a vector by a scalar, scalar product, vector product, unit vectors, rectangular unit vectors, spherical unit vectors.
- K2. Write the mathematical expressions for these theorems: divergence, Stokes', Green's.

Output Skills (Rule Application):

- R1. Given a scalar field determine its gradient and Laplacian in cartesian and spherical coordinates.
- R2. Given a vector field determine its divergence and curl in cartesian coordinates.

Output Skills (Problem Solving):

- S1. Given a table of vector identities involving differential operators, prove any given vector identity.
- S2. Solve problems involving basic vector algebra, including vector addition and subtraction, cross products and dot products.

External Resources (Required):

1. J. Reitz, F. Milford and R. Christy, *Foundations of Electromagnetic Theory*, 4th Edition, Addison-Wesley (1993).

THIS IS A DEVELOPMENTAL-STAGE PUBLICATION
OF PROJECT PHYSNET

The goal of our project is to assist a network of educators and scientists in transferring physics from one person to another. We support manuscript processing and distribution, along with communication and information systems. We also work with employers to identify basic scientific skills as well as physics topics that are needed in science and technology. A number of our publications are aimed at assisting users in acquiring such skills.

Our publications are designed: (i) to be updated quickly in response to field tests and new scientific developments; (ii) to be used in both classroom and professional settings; (iii) to show the prerequisite dependencies existing among the various chunks of physics knowledge and skill, as a guide both to mental organization and to use of the materials; and (iv) to be adapted quickly to specific user needs ranging from single-skill instruction to complete custom textbooks.

New authors, reviewers and field testers are welcome.

PROJECT STAFF

Andrew Schnepf	Webmaster
Eugene Kales	Graphics
Peter Signell	Project Director

ADVISORY COMMITTEE

D. Alan Bromley	Yale University
E. Leonard Jossem	The Ohio State University
A. A. Strassenburg	S. U. N. Y., Stony Brook

Views expressed in a module are those of the module author(s) and are not necessarily those of other project participants.

© 2001, Peter Signell for Project PHYSNET, Physics-Astronomy Bldg., Mich. State Univ., E. Lansing, MI 48824; (517) 355-3784. For our liberal use policies see:

<http://www.physnet.org/home/modules/license.html>.

REVIEW OF VECTOR ANALYSIS

by
R. D. Young

1. Introduction

You have already studied most of the mathematics needed for this course. However, there are probably some of the more formal vector identities (especially those involving the del operator, $\vec{\nabla}$) and vector theorems with which you are not too familiar. This unit will review vector analysis and refresh your memory regarding major vector theorems. The major vector theorems are summarized in Table 1-1. You should refer to this table whenever you meet a problem involving the differential operator, $\vec{\nabla}$.

1. Procedure

1. Read chapter 1 of the text (Reitz and Milford).
2. Underline in the text or write out the definitions and explanations of the terms in Output Skill K1.
3. Write down the various gradients, divergence, and curl called for in Output Skills R1 and R2. In particular,
 - gradient in rectangular coordinates (eq. 1-20)
 - gradient in spherical coordinates (eq. 1-23)
 - divergence in rectangular coordinates (eq. 1-31)
 - curl in rectangular coordinates (eq. 1-44)
4. Write down and memorize these theorems:
 - Divergence Theorem (p. 13 and eq. 1-37)
 - Stoke's Theorem (p. 16 and eq. 1-45)
 - Green's Theorem (eq. 1-57)
5. Solve the following problems in Reitz and Milford using Table 1-1 and 1-2 where appropriate:
 Problem 1-2, 1-16, 1-19, 1-20, 1-22, and 1-26.

Acknowledgments

The author would like to thank Illinois State University for support in the construction of this lesson. Preparation of this module was supported in part by the National Science Foundation, Division of Science Education Development and Research, through Grant #SED 74-20088 to Michigan State University.

